



Synthesis of novel diphosphite ligands derived from D-mannitol and their application in Cu-catalyzed enantioselective conjugate addition of organozinc to enones

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ABSTRACT

A series of novel chiral diphosphite ligands have been synthesized from D-mannitol derivatives and chlorophosphoric acid diary ester, and were successfully employed in the copper catalyzed enantioselective conjugate addition of organozinc reagents diethylzinc and dimethylzinc to cyclic and acyclic enones. The stereochemically matched combination of D-mannitol and (R)-H₈-binaphthyl in ligand 1,2:5,6-di-O-isopropylidene-3,4-bis[(R)-1,1'-H₈-binaphthyl-2,2'-diyl] phosphite-D-mannitol was essential to afford 93% ee for 3-ethylcyclohexanone, 92% ee for 3-ethylcyclopentanone, and 90% ee for 3-ethylcycloheptanone in toluene, using Cu(OTf)₂ as a catalytic precursor. The results clearly indicated that the chiral organocopper reagent exhibited high enantioselectivities for cyclic enones bearing different ring sizes. As for the backbone of this type of ligand, it has been demonstrated that 1,2:5,6-di-O-isopropylidene-D-mannitol was more efficient than 1,2:5,6-di-O-cyclohexylidene-D-mannitol. The sense of the enantiodiscrimination was mainly determined by the configuration of the diaryl phosphite moieties in the 1,4-addition of cyclic enones.

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1. Introduction

Asymmetric conjugate addition of organometallic reagents to α,β -unsaturated carbonyl compounds, including organozinc, -lithium, -aluminum, -boron, and Grignards is one of the most powerful tools for the stereoselective construction of carbon–carbon bonds in the synthesis of biologically active compounds,¹ such as erogorgiaene,² (–)-pumiliotoxin C,³ and β -D-mannosyl phosphomycoketide.⁴ During the past decades, much effort has been focused on the development of Cu-catalyzed enantioselective 1,4-conjugate additions of organozincs to α,β -unsaturated carbonyl compounds due to the low cost and a wide range of catalytic activities of copper salts in various reactions, such as oxidation, reduction, addition reactions to multiple bonds and cross coupling reactions. Since the pioneering discovery of Alexakis in 1993,⁵ dramatic breakthroughs have been achieved using numerous chiral phosphorus ligands, such as phosphoramidites,⁶ phosphites,⁷ P, N ligands⁸ and other types⁹ in this reaction. Of the various ligands mentioned above, phosphite ligands have been paid much attention recently, owing to their efficiency for 1,4-addition, facile preparation from a plethora of alcohols and chlorophosphites, and lower sensitivity to air and moisture. In this context, a great many

alcohols, derived from easily accessible and commercially available carbohydrates, have been used as starting materials for the synthesis of chiral phosphite ligands.¹⁰

D-Mannitol is one of the most important carbohydrates and is extremely useful for the synthesis of chiral ligands, since they are of high enantiomeric purity and are readily available. In addition, their highly functionalized property makes it straightforward to design novel ligands bearing specific structures through modifications. Some examples of this are classified according to D-mannitol skeletons and are schematically presented in Figure 1. As exhibited in Figure 1, versatile phosphorus ligands using D-mannitol as a backbone were synthesized, and applied in asymmetric catalytic reactions, such as the mono- and diphosphite ligands based on 1,4:3,6-di-O-dianhydro-D-mannitol, which gave excellent enantioselectivities in the Rh-catalyzed asymmetric hydrogenation of dimethyl itaconate and methyl N-acetamidoacrylate,¹¹ and diphosphite ligands based on 1,2:5,6-di-O-cyclohexylidene-D-mannitol afforded moderate enantioselectivities in Rh-catalyzed asymmetric hydroformylation.¹² A cooperative effect has been demonstrated between the stereogenic centers of the ligand backbone and the stereogenic binaphthyl phosphite moieties.

In our previous work, chiral diphosphite ligands **1a** and **1b** (Fig. 2) and P, N-ligands, based on 1,2:5,6-di-O-cyclohexylidene-D-mannitol, were synthesized and applied in the Cu-catalyzed conjugate addition of Et₂Zn to cyclic enones with moderate

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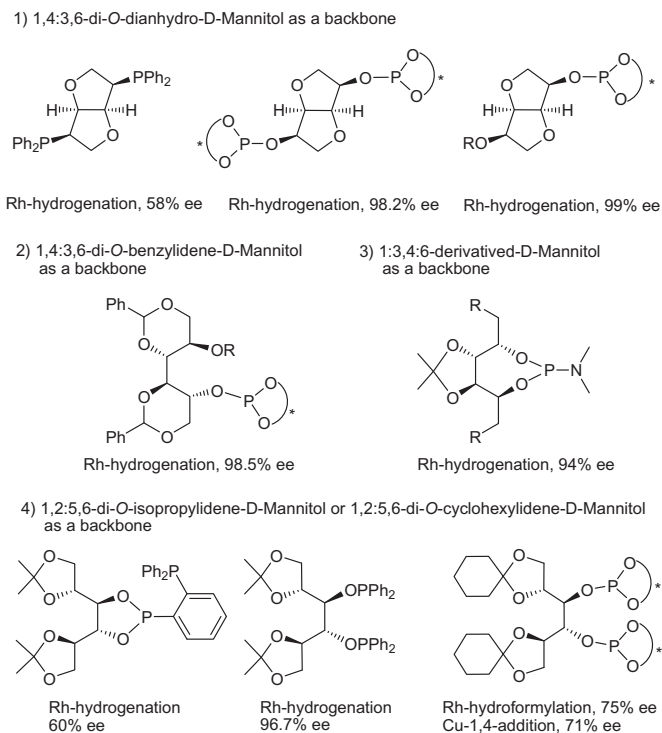


Figure 1. Selected phosphorus ligands based on D-mannitol derivatives.

enantioselectivity (71% ee). Two notable cooperative effects of the stereochemistry of the ligands on enantioselectivity were observed in this reaction, one between the phenylcarbamate substituent and the axially chiral binaphthyl moiety, another between the stereogenic centers of the mannitol and the chiral binaphthyl substituents. However, the reaction did have a major problem, in that the chiral cuprate-mediated reagent exhibited high substrate specificity.^{7a} As an effort to overcome the problem and enhance the enantioselectivity of the reaction, we considered modifying this type of ligand by introducing new substituents in to the D-mannitol skeleton and various diaryl phosphite moieties. Herein, we report the design and synthesis of novel ligands **1c–1d** and **2a–2d** (Fig. 2) based on D-mannitol, and their application in the Cu-catalyzed asymmetric 1,4-conjugate addition of dialkylzincs to cyclic and acyclic enones.

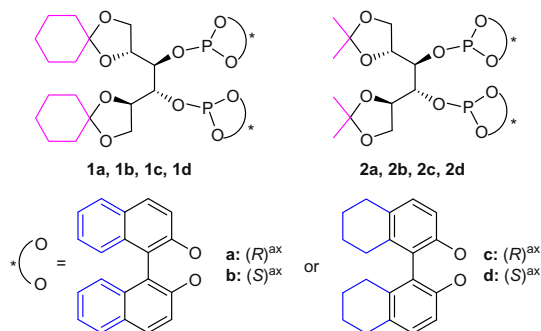


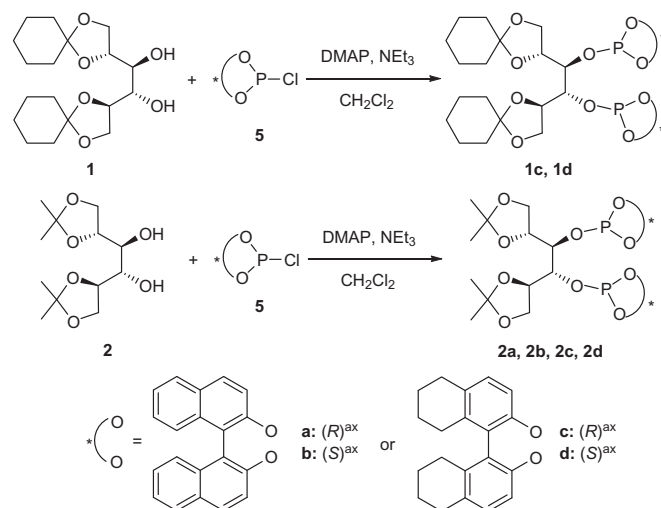
Figure 2. Chiral diphosphite ligands derived from D-mannitol.

2. Results and discussion

2.1. Synthesis of phosphite ligands 1c–1d and 2a–2d

As shown in Scheme 1, diphosphite ligands **1c–1d** and **2a–2d** were synthesized stereospecifically in one step from 1,2:5,6-di-O-

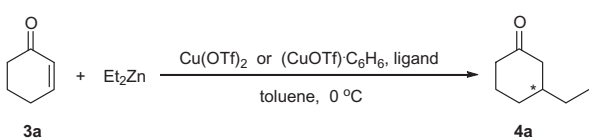
cyclohexylidene-D-mannitol **1** or 1,2:5,6-di-O-isopropylidene-D-mannitol **2**, and chlorophosphoric acid diary ester **5** derived from 2,2'-dihydroxy-5,5',6,6',7,7', 8,8'-octahydro-1,1'-binaphthol (H₈-binaphthol) or binaphthol. Compound **5** was conveniently prepared according to the literature procedures.¹³ In contrast with ligands **1c**, **1d**, **2c**, and **2d**, the binaphthyl moieties of ligands **1a**, **1b**, **2a**, and **2b** were replaced by H₈-binaphthyl (Fig. 2). All ligands were purified on a silica gel column under a nitrogen atmosphere with moderate to good yields. The white solid ligands **1c–1d** and **2a–2d** were air-stable at room temperature. The ³¹P NMR, ¹H NMR and ¹³C NMR were consistent with the expectation for these ligands.



Scheme 1. The synthesis of chiral diphosphite ligands derived from D-mannitol.

2.2. Asymmetric 1,4-conjugate addition of ZnR₂ to cyclic enones

With novel ligands **1c–d** and **2a–d** in hand, we evaluated their asymmetric induction abilities in the Cu-catalyzed asymmetric 1,4-addition of ZnEt₂ to 2-cyclohexenone **3a**, and the results are summarized in Table 1. In the first set of experiments, we tested, separately, the new ligands in toluene using Cu(OTf)₂ as a catalytic precursor (Table 1, entries 1–6). Ligand **1c** gave 77% yield and 75% ee (Table 1, entry 1). In contrast, the use of ligand **1d**, in which the carbohydrate skeleton was the same as in ligand **1c**, but the configuration of the H₈-binaphthyl moiety was opposite to that of **1c**, afforded similar yield (73%) but much lower enantioselectivity (47% ee) (Table 1, entry 2). Ligand **2a**, bearing 1,2:5,6-di-O-isopropylidene-D-mannitol as a backbone, produced 85% yield and 74% ee (Table 1, entry 3). Ligand **2b**, containing the same carbon skeleton as ligand **2a**, and opposite axial chirality of the binaphthyl moiety to **2a**, gave much lower yield and enantioselectivity (67% yield and 53% ee) (Table 1, entry 4). Using ligand **2c**, the enantioselectivity of the reaction was enhanced to 83% (Table 1, entries 3 and 5). However, employing **2d** in the reaction, the enantioselectivity of the reaction decreased to 45% (Table 1, entries 4 and 6). The comparison of these results clearly indicated that enantioselectivities depended strongly on the steric and electronic properties of the C-1, C-2, C-5, and C-6 stereocenter, and the mannitol skeleton and that the axially chiral binaphthyl or H₈-binaphthyl moiety had a synergic effect on the enantioselectivity of the reaction. Thus, for the chiral skeleton of this type of ligands, 1,2:5,6-di-O-isopropylidene-D-mannitol was more efficient than 1,2:5,6-di-O-cyclohexylidene-D-mannitol to afford higher enantioselectivity in toluene using Cu(OTf)₂ as a catalytic precursor (Table 1, entries 1, 2, 5, and 6).

Table 1
Cu-catalyzed enantioselective 1,4-addition of ZnEt₂ to 2-cyclohexenone^a


Entry	Cu salts	L.	L/Cu	Conv. ^b (%)	Y. ^b (%)	%ee (Conf.) ^c
1	Cu(OTf) ₂	1c	2	>99	77	75 (R)
2	Cu(OTf) ₂	1d	2	99	73	47 (S)
3	Cu(OTf) ₂	2a	2	>99	85	74 (R)
4	Cu(OTf) ₂	2b	2	99	67	53 (S)
5	Cu(OTf) ₂	2c	2	>99	90	83 (R)
6	Cu(OTf) ₂	2d	2	98	83	45 (S)
7	(CuOTf) ₂ ·C ₆ H ₆	2c	2	>99	77	77 (R)
8	Cu(OTf) ₂	2c	0.5	99	85	65 (R)
9	Cu(OTf) ₂	2c	1.0	99	92	68 (R)
10	Cu(OTf) ₂	2c	1.1	>99	98	80 (R)
11	Cu(OTf) ₂	2c	3.0	>99	87	79 (R)

^a Reaction conditions: Cu(OTf)₂ (0.005 mmol) or (CuOTf)₂·C₆H₆ (0.0025 mmol), ligand (0.01 mmol), ZnEt₂ (1.0 M in hexane, 0.6 mmol), **3a** (0.25 mmol), toluene (4 mL), 0 °C, 4 h.

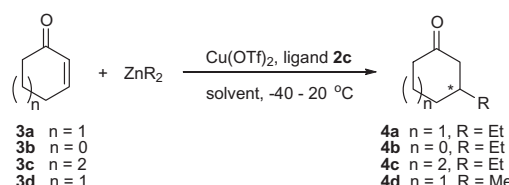
^b The data on conversion and yield were determined by GC using dodecane as an internal standard with a SE-30 column (30 m × 0.32 mm I.D.).

^c The enantiomeric excess of **4a** was determined by GC equipped with a Chiral-dex A-TA column (50 m × 0.25 mm I.D.). The absolute configuration of **4a** was determined by comparison with an authentic sample.

We next examined the influence of the copper precursor as well as copper/ligand ratio on the reactive activity and enantioselectivity (Table 1, entries 6–9). In comparison to Cu(OTf)₂, (CuOTf)₂·C₆H₆ as a catalytic precursor could enhance the enantioselectivity in the presence of ligand **1a**.^{7a,7b} However, the catalyst formed in situ by (CuOTf)₂·C₆H₆ and ligand **2c** gave slightly lower enantioselectivity (77% ee) (Table 1, entry 7). Thus, Cu(OTf)₂, owing to its lower sensitivity to air and moisture, was chosen as a catalytic precursor for the next experiments. An enhancement in enantioselectivity was observed with a ratio of Cu(OTf)₂/ligand ranging from 1/0.5 to 1/2 (Table 1, entries 5, 8–10). A further increase of the ratio of ligand to Cu(OTf)₂ brought no obvious change of the catalytic activity or enantioselectivity (Table 1, entry 11).

With ligand **2c** being identified as the most effective ligand, using Cu(OTf)₂ as a catalyst precursor, the conjugate addition of ZnEt₂ to **3a** in different solvents and at various temperatures was carried out separately. A profound solvent effect on this reaction was observed (Table 2, entries 1–3). The reactions in solvents CH₂Cl₂ and Et₂O, respectively, gave lower enantioselectivities (Table 2, entries 1 and 2). When the reaction was carried out in THF, low catalytic activity and moderate enantioselectivity was obtained (Table 2, entry 3). Toluene was proven to be an appropriate solvent in our cases (Table 1, entry 5 vs Table 2, entries 1–3). Performing the reaction in toluene at –20 °C allowed the highest enantioselectivity with up to 93% ee (Table 2, entry 6). In contrast, at –10 °C, –30 °C, and –40 °C, respectively, no significant improvement of the enantioselectivities was observed (Table 2, entries 4–8).

Under the optimal reaction conditions, a survey of the reaction scope was then undertaken. When ZnMe₂ was used in the reaction instead of ZnEt₂, the enantioselectivity of the corresponding 1,4-adduct decreased to 69% ee (Table 2, entry 9). A similar result in that ZnMe₂ gave lower enantioselectivity than ZnEt₂ was also observed by Mauduit et al. They found that 90% ee and 45% ee were obtained in the 1,4-addition of ZnEt₂ or ZnMe₂ to 2-cyclohexenone, respectively.¹⁴ A possible pathway for the 1,4-addition had been proposed by Feringa et al. The alkyl transfer to the β-position of the enone in complex **I** could produce alkylzinc enolate **II**, which upon protonation generated 1,4-adduct **III** (Scheme 2).^{6f} Based on the mechanism, we presumed that the ethyl group performed a more suitable steric and electronic effect than a methyl when the alkyl transferred to the enone. Thus, we still select ZnEt₂ as an organozinc reagent in the next entries. The asymmetric conjugate addition reactions of ZnEt₂ to 2-cyclopentenone **3b** and 2-cycloheptenone **3c** as the substrate proceeded successfully to afford 92% ee and 90% ee of the products, respectively (Table 2, entries 10 and 11). It was demonstrated that ligand **2c** was efficient for the Cu-catalyzed asymmetric conjugate addition of ZnEt₂ to cyclic enones bearing different ring sizes. The results were consistent with Chan et al.'s observation that three cyclic enones **3a–3c** were all converted smoothly to the 1,4-adducts with high enantiomeric excesses.^{7c} In addition, the asymmetric 1,4-additions of ZnEt₂ to acyclic enones were tested. Unfortunately, low enantioselectivities were observed: 18% ee for 1,3-diphenyl-2-propenone **3e**, 45% ee

Table 2
Cu-catalyzed enantioselective 1,4-addition of organozinc to cyclic enones^a


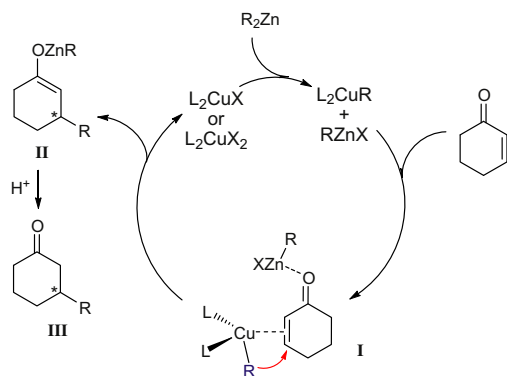
Entry	Sub.	Prod.	Soln	T (°C)	t (h)	Conv. ^b (%)	Y. ^b (%)	%ee. (Conf.) ^b
1	3a	4a	CH ₂ Cl ₂	0	4	78	40	2 (R)
2	3a	4a	Et ₂ O	0	4	99	77	16 (R)
3	3a	4a	THF	0	4	45	43	66 (R)
4	3a	4a	Toluene	20	4	>99	99	64 (R)
5	3a	4a	Toluene	–10	12	>99	>99	89 (R)
6	3a	4a	Toluene	–20	12	>99	>99	93 (R)
7	3a	4a	Toluene	–30	12	>99	98	92 (R)
8	3a	4a	Toluene	–40	12	>99	>99	88 (R)
9 ^{c,d}	3d	4d	Toluene	–20	20	>99	55	69 (R)
10	3b	4b	Toluene	–20	12	97	59	92 (R)
11 ^c	3c	4c	Toluene	–20	12	99	97	90 (R)

^a Reaction conditions: Cu(OTf)₂ (0.005 mmol), ligand **2c** (0.01 mmol), ZnEt₂ (1.0 M in hexane, 0.6 mmol), enone (0.25 mmol), solvent (4 mL).

^b The date on conversion, yield, enantiomeric excess, and the absolute configuration of the products were determined using the same conditions as noted in Table 1.

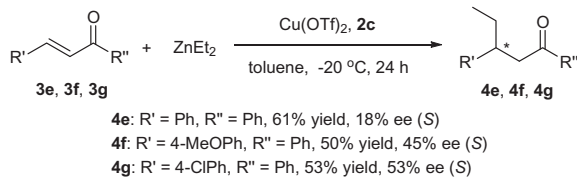
^c The enantiomeric excess was determined by GC equipped with a CP-Chirasil-Dex CB (25 m × 0.25 mm I.D.).

^d ZnMe₂ (1.1 M in toluene, 0.6 mmol), 24 h.



Scheme 2. Postulated mechanism of the Cu-catalyzed asymmetric addition reaction.^{6f}

for 3-(4-methoxyphenyl)-1-phenyl-2-propenone **3f**, and 53% ee for 3-(4-chlorophenyl)-1-phenyl-2-propenone **3g** (Scheme 3). The results indicated that the newly developed diphosphite ligand **2c** was inferior for these acyclic enones. In the 1,4-addition of cyclic enones, the sense of the enantiodiscrimination was mainly determined by the configuration of the binaphthyl moieties of the ligand.



Scheme 3. Cu-catalyzed 1,4-conjugate addition of ZnEt₂ to acyclic enones.

3. Conclusion

In conclusion, we have synthesized a novel series of diphosphite ligands, which are derived from 1,2:5,6-di-*O*-isopropylidene-*D*-mannitol and 1,2:5,6-di-*O*-cyclohexylidene-*D*-mannitol. The ligands were successfully applied in the copper-catalyzed asymmetric conjugate addition of dialkylzincs to cyclic enones bearing different ring sizes with high enantioselectivities. The stereogenic centers of the *D*-mannitol skeleton and the axially chiral diaryl moieties of ligands had a synergic effect on the reactive activity and enantioselectivity of the reaction. The sense of the enantiodiscrimination was predominantly controlled by the configuration of the biaryl moieties of the ligand in the 1,4-addition of cyclic enones. The application of these types of ligands to other transition metal-catalyzed asymmetric reactions is ongoing in our laboratory.

4. Experimental

4.1. General

Melting points were determined using an X-4 melting point apparatus and were uncorrected. Optical rotations were measured on a Perkin–Elmer 241 MC polarimeter at 20 °C. NMR spectra were recorded on Bulker 300 MHz or Bulker 400 MHz spectrometers. ¹H NMR and ¹³C NMR spectra were reported in parts per million with TMS (δ = 0.00 ppm) as an internal standard. ³¹P NMR spectra were reported in parts per million with 85% H₃PO₄ as an external reference. Proton chemical shifts (δ) and coupling constants (*J*) were reported in ppm and in Hertz, respectively. Spin multiplicities were

given as s (singlet), d (doublet), t (triplet) and m (multiplet). HRMS were recorded on a Bulker micrOTOF-QII mass instrument.

All non-aqueous reactions and manipulations were performed under an N₂ atmosphere with standard Schlenk techniques. Reactions were monitored by thin layer chromatography (TLC, silica gel GF254 plates). Column chromatography separations were conducted on silica gel (200–300 mesh). NEt₃, THF, Et₂O, and toluene were distilled with Na and benzophenone as an indicator, and CH₂Cl₂ was dried over CaH₂ before use. H₈-binaphthol was prepared according to a literature procedure.^{13a} All the other chemicals were obtained commercially and used without further purification.

4.2. Synthesis of ligands 1c–1d and 2a–2d

4.2.1. 1,2:5,6-Di-*O*-cyclohexylidene-3,4-bis[(*R*)-1,1'-H₈-binaphthyl-2,2'-diyl]phosphite-*D*-mannitol 1c

To a 100 mL Schlenk flask equipped with a condenser were added 1.77 g of (*R*)-H₈-binaphthol, 20 mL of toluene, and 10 mL of PCl₃. Then the mixture was refluxed for 22 h under a nitrogen atmosphere. After removal of the excessive PCl₃ and toluene, the residue was dissolved using 20 mL of toluene, and then was transferred to another Schlenk flask, and toluene was removed in vacuo to obtain the compound (*R*)-1,1'-H₈-binaphthyl-2,2'-diyl-chlorophosphite **5c** as a white powder, which was used directly in the following step without further purification. To a stirred solution of 1,2:5,6-di-*O*-cyclohexylidene-*D*-mannitol **1** (206 mg, 0.6 mmol), **5c** (646 mg, 1.8 mmol), and DMAP (14.6 mg, 0.12 mmol) in CH₂Cl₂ (20 mL) at –10 °C, NEt₃ (0.34 mL) was slowly added using a syringe over 2 min, and the solution was stirred for 0.5 h at –10 °C. The mixture was then stirred at rt for 1 h. CH₂Cl₂ was distilled off in vacuo, and then toluene (20 mL) was added. The solid was removed by filtration through a pad of silica gel, and the solvent was removed under reduced pressure. The residue was purified by flash chromatography (toluene, *R*_f = 0.56), and furnished ligand **1c** as a white foamy solid (266 mg, 45% yield), mp 128–130 °C. $[\alpha]_D^{20}$ = –165 (c 0.2, CH₂Cl₂). ³¹P NMR (162 MHz, DMSO-*d*₆): δ 146.83 ppm. ¹H NMR (400 MHz, DMSO-*d*₆): δ 7.24 (d, *J* = 7.6 Hz, 2H), 7.05 (d, *J* = 8.0 Hz, 2H), 6.94 (d, *J* = 8.0 Hz, 2H), 6.85 (d, *J* = 8.0 Hz, 2H), 4.53 (t, *J* = 6.4 Hz, 2H), 4.16 (d, *J* = 5.2 Hz, 2H), 4.01 (t, *J* = 7.2 Hz, 2H), 3.77 (t, *J* = 6.8 Hz, 2H), 2.76 (m, 8H), 2.591 (m, 4H), 2.12 (m, 4H), 1.34–1.71 (m, 36H) ppm. ¹³C NMR (100 MHz, DMSO-*d*₆): δ 145.40, 137.98, 137.01, 134.65, 133.76, 129.45, 129.18, 128.77, 127.03, 118.63, 118.39, 109.06, 75.69, 73.58, 64.97, 35.57, 33.75, 28.40, 27.25, 25.17, 24.63, 23.59, 23.41, 21.94, 21.84 ppm. HRMS (ESI) calcd for: C₅₈H₆₈NaO₁₀P₂ (M+Na)⁺ 1009.4180, found: 1009.4176, 0.4 ppm.

4.2.2. 1,2:5,6-Di-*O*-cyclohexylidene-3,4-bis[(*S*)-1,1'-H₈-binaphthyl-2,2'-diyl]phosphite-*D*-mannitol 1d

(*S*)-1,1'-H₈-Binaphthyl-2,2'-diyl-chlorophosphite **5d** was synthesized by the same procedure as that of **5c**, and was used directly without further purification. Treatment of compound **1** (206 mg, 0.6 mmol), **5d**, (646 mg, 1.8 mmol) and DMAP (14.6 mg, 0.12 mmol) as described for the synthesis of ligand **1c** afforded ligand **1d**, which was purified by flash chromatography (toluene, *R*_f = 0.65) to produce a white solid (252 mg, 43% yield), mp 111–112 °C. $[\alpha]_D^{20}$ = +184 (c 0.2, CH₂Cl₂). ³¹P NMR (162 MHz, DMSO-*d*₆): δ 144.91 ppm. ¹H NMR (400 MHz, DMSO-*d*₆): δ 7.15 (d, *J* = 8.4 Hz, 2H), 7.08 (d, *J* = 8.0 Hz, 2H), 7.01 (d, *J* = 8.4 Hz, 2H), 6.86 (d, *J* = 8.4 Hz, 2H), 4.44 (t, *J* = 8.8 Hz, 2H), 4.21 (dd, *J* = 12.8, 5.6 Hz, 2H), 3.87 (dd, *J* = 8.8, 6.4 Hz, 2H), 3.76 (dd, *J* = 8.4, 4.8 Hz, 2H), 2.78 (m, 8H), 2.60 (m, 4H), 2.13 (m, 4H), 1.73 (m, 12H), 1.56–1.28 (m, 24H) ppm. ¹³C NMR (100 MHz, DMSO-*d*₆): δ 145.28, 137.90, 134.67, 133.67, 131.48, 129.47, 129.00, 128.63, 128.59, 127.06, 118.75, 118.58, 109.34, 75.06, 73.13, 67.30, 65.48,

35.94, 34.17, 29.70, 28.35, 27.18, 27.11, 24.45, 23.55, 23.15, 22.30, 21.89, 21.83, 21.78 ppm. HRMS (ESI) calcd for: $C_{58}H_{68}NaO_{10}P_2$ (M+Na)⁺ 1009.4180, found: 1009.4167, 1.3 ppm.

4.2.3. 1,2:5,6-Di-O-isopropylidene-3,4-bis[(R)-1,1'-binaphthyl-2,2'-diyl]phosphite-D-mannitol 2a

(R)-1,1'-Binaphthyl-2,2'-diyl-chlorophosphite **5a** was synthesized by the same procedure as that of **5c**, and was used directly without further purification. Treatment of 1,2:5,6-di-O-isopropylidene-D-mannitol **2** (79 mg, 0.3 mmol), **5a** (421 mg, 1.2 mmol) and DMAP (7.3 mg, 0.06 mmol) as described for the synthesis of ligand **1c** afforded ligand **2a**, which was purified by flash chromatography (toluene, R_f = 0.36) to afford a white solid (200 mg, 75% yield), mp 119–121 °C. $[\alpha]_D^{20}$ = −309 (c 0.2, CH_2Cl_2). ³¹P NMR (162 MHz, DMSO- d_6): δ 153.30 ppm. ¹H NMR (300 MHz, DMSO- d_6): δ 8.15 (d, J = 8.7 Hz, 2H), 8.10 (d, J = 8.1 Hz, 2H), 8.07 (d, J = 8.1 Hz, 2H), 7.96 (d, J = 9.0 Hz, 2H), 7.56 (d, J = 9.0 Hz, 2H), 7.52 (t, J = 7.8 Hz, 4H), 7.43 (d, J = 9.0 Hz, 2H), 7.36 (t, J = 8.1 Hz, 4H), 7.23 (m, 4H), 4.83 (m, 2H), 4.42 (dd, J = 10.5, 5.7 Hz, 2H), 4.00 (t, J = 7.8 Hz, 2H), 3.77 (m, 2H), 1.49 (s, 6H), 1.35 (s, 6H) ppm. ¹³C NMR (75 MHz, DMSO- d_6): δ 147.45, 147.39, 146.53, 131.97, 131.56, 131.15, 130.76, 130.57, 130.03, 128.58, 126.68, 126.50, 126.00, 125.89, 125.32, 125.09, 123.57, 123.50, 121.76, 121.56, 121.37, 108.28, 76.06, 75.72, 74.28, 64.44, 25.71, 24.28 ppm. HRMS (ESI) calcd for: $C_{52}H_{44}NaO_{10}P_2$ (M+Na)⁺ 913.2302, found: 913.2300, 0.2 ppm.

4.2.4. 1,2:5,6-Di-O-isopropylidene-3,4-bis[(S)-1,1'-binaphthyl-2,2'-diyl]phosphite-D-mannitol 2b

(S)-1,1'-Binaphthyl-2,2'-diyl-chlorophosphite **5b** was prepared using the same procedure as **5c**, and was used directly without further purification. Treatment of 1,2:5,6-di-O-isopropylidene-D-mannitol **2** (157 mg, 0.6 mmol), **5b** (505 mg, 1.44 mmol) and DMAP (14.6 mg, 0.12 mmol) as described for the synthesis of ligand **1c** afforded ligand **2b**, which was purified by flash chromatography (toluene, R_f = 0.21) to obtain a white solid (221 mg, 40% yield), mp 115–117 °C. $[\alpha]_D^{20}$ = +496 (c 0.2, CH_2Cl_2). ³¹P NMR (162 MHz, DMSO- d_6): δ 150.41 ppm. ¹H NMR (400 MHz, DMSO- d_6): δ 8.11 (t, J = 8.8 Hz, 4H), 8.03 (d, J = 8.0 Hz, 2H), 7.90 (d, J = 8.8 Hz, 2H), 7.55 (m, 6H), 7.45 (d, J = 8.8 Hz, 2H), 7.40 (d, J = 8.0 Hz, 2H), 7.36 (d, J = 7.6 Hz, 2H), 7.27 (d, J = 8.0 Hz, 2H), 7.20 (d, J = 7.2 Hz, 2H), 4.63 (t, J = 8.8 Hz, 2H), 4.42 (dd, J = 12.8, 5.6 Hz, 2H), 4.06 (dd, J = 8.8, 6.4 Hz, 2H), 3.99 (dd, J = 8.8, 4.8 Hz, 2H), 1.40 (s, 6H), 1.26 (s, 6H) ppm. ¹³C NMR (100 MHz, DMSO- d_6): δ 147.39, 147.34, 146.42, 131.98, 131.70, 131.22, 130.86, 130.65, 130.02, 128.67, 128.60, 126.74, 126.60, 126.01, 125.39, 125.18, 123.54, 123.49, 121.75, 121.66, 121.59, 109.12, 75.17, 75.07, 73.42, 65.85, 26.50, 24.89 ppm. HRMS (ESI) calcd for: $C_{52}H_{44}NaO_{10}P_2$ (M+Na)⁺ 913.2302, found: 913.2280, 1.5 ppm.

4.2.5. 1,2:5,6-Di-O-isopropylidene-3,4-bis[(R)-1,1'-H₈-binaphthyl-2,2'-diyl]phosphite-D-mannitol 2c

Treatment of 1,2:5,6-di-O-isopropylidene-D-mannitol **2** (157 mg, 0.6 mmol), **5c** (860 mg, 2.4 mmol) and DMAP (14.6 mg, 0.12 mmol) as described for the synthesis of ligand **1c** afforded ligand **2c**, which was purified by flash chromatography (toluene, R_f = 0.32) to obtain a white solid (273 mg, 50% yield), mp 132–133 °C. $[\alpha]_D^{20}$ = −205 (c 0.2, CH_2Cl_2). ³¹P NMR (162 MHz, DMSO- d_6): δ 146.52 ppm. ¹H NMR (400 MHz, DMSO- d_6): δ 7.13 (d, J = 8.4 Hz, 2H), 7.05 (d, J = 8.4 Hz, 2H), 6.96 (d, J = 8.0 Hz, 2H), 6.88 (d, J = 8.0 Hz, 2H), 4.55 (dd, J = 8.8, 5.6 Hz, 2H), 4.19 (dd, J = 11.6, 5.6 Hz, 2H), 3.95 (t, J = 8.0 Hz, 2H), 3.73 (dd, J = 8.8, 5.6 Hz, 2H), 2.77 (m, 8H), 2.58–2.63 (m, 4H), 2.08–2.16 (m, 4H), 1.72 (m, 12H), 1.47 (m, 4H), 1.42 (s, 6H), 1.31 (s, 6H) ppm. ¹³C NMR (100 MHz, DMSO- d_6): δ 145.48, 145.40, 137.95, 136.99, 134.63, 133.74, 129.43, 129.13, 128.76, 128.71, 127.03, 118.73, 118.44, 108.41, 75.55, 75.30, 74.02, 64.96, 28.40, 27.25, 27.17,

25.98, 24.47, 21.96, 21.84 ppm. HRMS (ESI) calcd for: $C_{52}H_{60}NaO_{10}P_2$ (M+Na)⁺ 929.3554, found: 929.3559, 0.5 ppm.

4.2.6. 1,2:5,6-Di-O-isopropylidene-3,4-bis[(S)-1,1'-H₈-binaphthyl-2,2'-diyl]phosphite-D-mannitol 2d

Treatment of 1,2:5,6-di-O-isopropylidene-D-mannitol **2** (157 mg, 0.6 mmol), **5d** (517 mg, 1.44 mmol) and DMAP (14.6 mg, 0.12 mmol) as described for the synthesis of ligand **1c** afforded ligand **2d**, which was purified by flash chromatography (toluene, R_f = 0.26) to obtain a white solid (229 mg, 42% yield), mp 118–120 °C. $[\alpha]_D^{20}$ = +213 (c 0.2, CH_2Cl_2). ³¹P NMR (162 MHz, DMSO- d_6): δ 144.38 ppm. ¹H NMR (400 MHz, DMSO- d_6): δ 7.15 (d, J = 8.8 Hz, 2H), 7.07 (d, J = 8.0 Hz, 2H), 7.02 (d, J = 8.0 Hz, 2H), 6.87 (d, J = 8.4 Hz, 2H), 4.44 (t, J = 8.4 Hz, 2H), 4.22 (dd, J = 12.8, 5.6 Hz, 2H), 3.89 (dd, J = 8.4, 6.0 Hz, 2H), 3.77 (dd, J = 8.8, 5.2 Hz, 2H), 2.78 (m, 8H), 2.61 (m, 4H), 2.08–2.17 (m, 4H), 1.73 (m, 12H), 1.47 (m, 4H), 1.37 (s, 6H), 1.28 (s, 6H) ppm. ¹³C NMR (100 MHz, DMSO- d_6): δ 145.36, 145.31, 137.95, 137.07, 134.74, 133.69, 129.50, 129.02, 128.71, 128.67, 127.11, 118.86, 118.65, 108.93, 74.87, 74.76, 73.49, 65.76, 28.40, 27.25, 27.18, 26.48, 25.00, 21.96, 21.90, 21.84 ppm. HRMS (ESI) calcd for: $C_{52}H_{60}NaO_{10}P_2$ (M+Na)⁺ 929.3554, found: 929.3552, 0.2 ppm.

4.3. Representative procedure for the asymmetric 1,4-addition of organozincs to 2-cyclohexenone 3a

A solution of Cu(OTf)₂ (0.005 mmol, 1.8 mg) and ligand **2c** (0.01 mmol, 9.1 mg) in toluene (4 mL) was stirred for 1 h at rt under nitrogen. After the solution was cooled down to 0 °C, 2-cyclohexenone (0.25 mmol, 0.024 mL) was added and the solution was stirred for 10 min at 0 °C, then ZnEt₂ (0.6 mmol, 0.6 mL of 1.0 M solution in hexane) was added dropwise using a syringe over 2 min. After 4 h, the reaction was quenched by H₂O (2 mL) and 2 M HCl (2 mL), and extracted with ethyl acetate (5 mL × 3). The combined extracts were washed using saturated NaHCO₃ solution, brine, and then dried over anhydrous Na₂SO₄, filtered, and concentrated to afford the crude product. The conversion and the yield were determined by a GC equipped with a SE-30 column (30 m × 0.32 mm I.D.) using dodecane as an internal standard. The enantiomeric excess was determined by GC with a Chiraldex A-TA column (50 m × 0.25 mm I.D.). The absolute configuration was determined by comparison with authentic samples. The analytic conditions for the 1,4-adducts are as follows.

4.3.1. 3-Ethylcyclohexanone 4a

Chiraldex A-TA column (50 m × 0.25 mm I.D.) at 120 °C constant. t_R = 11.52 min for enantiomer (R), and t_R = 11.79 min for enantiomer (S).

4.3.2. 3-Ethylcyclopentanone 4b

Chiraldex A-TA column (50 m × 0.25 mm I.D.) at 110 °C constant. t_R = 12.76 min for enantiomer (R), and t_R = 12.46 min for enantiomer (S).

4.3.3. 3-Ethylcycloheptanone 4c

CP-Chirasil-Dex CB (25 m × 0.25 mm I.D.) at 120 °C constant. t_R = 13.70 min for enantiomer (R), and t_R = 13.31 min for enantiomer (S).

4.3.4. 3-Methylcyclohexanone 4d

CP-Chirasil-Dex CB (25 m × 0.25 mm I.D.) at 70 °C constant. t_R = 37.84 min for enantiomer (R), and t_R = 38.69 min for enantiomer (S).

4.3.5. 1, 3-Diphenyl-1-pentanone 4e

Daicel Chiralcel AD-H 25 cm × 4.6 mm I.D., hexane/*i*-PrOH = 95/5, flow rate = 0.5 mL/min at 25 °C, detected at 254 nm. t_R = 15.99 min for enantiomer (R), and t_R = 13.40 min for enantiomer (S).

4.3.6. 3-(4-Methoxyphenyl)-1-phenylpentanone 4f

Daicel Chiralcel AD-H 25 cm $\times$ 4.6 mm I.D., hexane/*i*-PrOH = 95/5, flow rate = 0.5 mL/min at 25 °C, detected at 254 nm. t_R = 25.22 min for enantiomer (*R*), and t_R = 17.80 min for enantiomer (*S*).

4.3.7. 3-(4-Chlorophenyl)-1-phenylpentanone 4g

Daicel Chiralcel AD-H 25 cm $\times$ 4.6 mm I.D., hexane/*i*-PrOH = 95/5, flow rate = 0.5 mL/min at 25 °C, detected at 254 nm. t_R = 18.90 min for enantiomer (*R*), and t_R = 14.95 min for enantiomer (*S*).

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